gpeyt1-26 · Flow-net seepage

GEO PE · Long Calculation Problems I — Topic 6: Groundwater & Seepage

Problem

A concrete diversion weir on a silty sand foundation (k = 2.5e-04 cm/s, isotropic) has headwater 34 ft and tailwater 6 ft above the foundation surface. A properly drawn flow net beneath the weir has Nf = 4 flow channels and Nd = 14 equipotential drops. The weir is 450 ft long. The sheet-pile cutoff is 20 ft deep and the sand is 45 ft thick.

Most nearly, the total seepage under the weir is:

A. 3,098 ft³/day
B. 31,252 ft³/day
C. 2,551 ft³/day
D. 6 ft³/day

Given equations / data

> q = k H Nf/Nd per unit length; H = total head loss; 1 cm/s = 2,835 ft/day

Worked solution

1. k = 2.5e-04 cm/s × 2,835 = 0.709 ft/day
2. H = 34 − 6 = 28 ft
3. q = 0.709 × 28 × 4/14 = 5.67 ft³/day/ft
4. Q = 5.67 × 450 = 2,551 ft³/day

Final answer

2,551 ft³/day

Why the other values are wrong

- **3,098 ft³/day** — used headwater depth instead of head difference
- **31,252 ft³/day** — inverted Nf/Nd
- **6 ft³/day** — reported per-foot flow (forgot length)

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.16.4 Flow Nets, p. 190; §3.16.1 Darcy's Law

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=RqaahrVtV4k, https://www.youtube.com/watch?v=ng3In5ayyaE, https://www.youtube.com/watch?v=_qeKY4icSPs (no creator values copied).

Personal use only — draft study material for Jayce Wright. Not for redistribution. Not reviewed by a licensed PE.

gpeyt1-27 · Exit gradient / boiling

GEO PE · Long Calculation Problems I — Topic 6: Groundwater & Seepage

Problem

Inside a sheet-pile cofferdam excavated in fine sand (γsat = 124 pcf, Gs = 2.66, e = 0.62), a flow net shows the last equipotential square adjacent to the excavation floor has a head loss of Δh = 2.8 ft over a flow-path length of 4.5 ft. Total head difference across the cofferdam is 16 ft and the sheet piles are embedded 14 ft below the floor.

Most nearly, the factor of safety against boiling at the exit is:

A. 3.19
B. 1.59
C. 0.63
D. 0.86

Given equations / data

> icr = γ′/γw; iexit = Δh/l; FS = icr/iexit; γw = 62.4 pcf

Worked solution

1. γ′ = 124 − 62.4 = 61.6 pcf → icr = 0.987
2. (check: (Gs−1)/(1+e) = 1.025)
3. iexit = 2.8/4.5 = 0.622
4. FS = 0.987/0.622 = 1.59

Final answer

1.59

Why the other values are wrong

- **3.19** — used γsat/γw for icr
- **0.63** — inverted FS
- **0.86** — used total head/embedment as the exit gradient

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.16.3 Flow Through Soil / §3.16.4 Flow Nets, pp. 188–190

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=RqaahrVtV4k (no creator values copied).

Personal use only — draft study material for Jayce Wright. Not for redistribution. Not reviewed by a licensed PE.

gpeyt1-28 · Unconfined well (Thiem)

GEO PE · Long Calculation Problems I — Topic 6: Groundwater & Seepage

Problem

A fully penetrating dewatering well in an unconfined sand-and-gravel aquifer (K = 45 ft/day) resting on horizontal impermeable clay is pumped to steady state. Observation wells at r1 = 25 ft and r2 = 150 ft show saturated thicknesses (water level above the clay) of h1 = 38.0 ft and h2 = 41.5 ft. The well radius is 0.5 ft, the static saturated thickness was 44 ft, and the pump is set 5 ft above the clay.

Most nearly, the pumping rate is:

A. 50,551 ft³/day
B. 21,954 ft³/day
C. 967 ft³/day
D. 43,908 ft³/day

Given equations / data

> Unconfined (Dupuit–Thiem): $Q = \pi K(h2^2 - h1^2)/\ln(r2/r1)$

Worked solution

1. $h2^2 - h1^2 = 1722.25 - 1444.00 = 278.25\ ft^2$
2. $\ln(150/25) = 1.7918$
3. $Q = \pi \times 45 \times 278.25/1.7918 = 21{,}954\ ft^3/day\ (\approx 114\ gpm)$

Final answer

21,954 ft³/day

Why the other values are wrong

- **50,551 ft³/day** — used log10
- **967 ft³/day** — squared the difference instead of differencing squares
- **43,908 ft³/day** — used 2π (confined form)

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §6.6.4 Well Analysis–Steady State (unconfined aquifer); §3.16.5 Construction Dewatering

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=RqaahrVtV4k (no creator values copied).

Personal use only — draft study material for Jayce Wright. Not for redistribution. Not reviewed by a licensed PE.

gpeyt1-29 · Confined aquifer K

GEO PE · Long Calculation Problems I — Topic 6: Groundwater & Seepage

Problem

A pumping test in a 30 ft thick confined sand aquifer (overlain by 22 ft of clay) runs at a steady 350 gpm. Drawdowns: 9.2 ft at r = 40 ft and 5.1 ft at r = 200 ft. The pumping well radius is 0.67 ft; the piezometric surface is 8 ft below grade before pumping.

Most nearly, the hydraulic conductivity K is:

A. 140 ft/day
B. 61 ft/day
C. 40 ft/day
D. 281 ft/day

Given equations / data

> Confined: Q = 2πKb(h2 − h1)/ln(r2/r1); h2 − h1 = s1 − s2; 1 ft³ = 7.48 gal

Worked solution

1. Q = 350 gpm / 7.48 × 1,440 = 67,380 ft³/day
2. h2 − h1 = 9.2 − 5.1 = 4.1 ft
3. K = 67,380 × ln(200/40) / (2π × 30 × 4.1) = 140.3 ft/day

Final answer

140 ft/day

Why the other values are wrong

- **61 ft/day** — used log10
- **40 ft/day** — added drawdowns
- **281 ft/day** — used π instead of 2π

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §6.6.4 Well Analysis–Steady State (confined aquifer); UFC 3-220-05 Ch. 4

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=RqaahrVtV4k (no creator values copied).

Personal use only — draft study material for Jayce Wright. Not for redistribution. Not reviewed by a licensed PE.

gpeyt1-30 · Layered k (vertical)

GEO PE · Long Calculation Problems I — Topic 6: Groundwater & Seepage

Problem

A 12 ft soil column beneath a detention pond consists of 4 ft of sand (k = 2×10^{-3} cm/s), 2 ft of silty clay (k = 5×10^{-6} cm/s), and 6 ft of silty sand (k = 8×10^{-4} cm/s), top to bottom. The total head loss across the column is 9 ft. Flow is vertical (downward).

Most nearly, the equivalent vertical hydraulic conductivity is:

A. 1.1e-03 cm/s
B. 2.9e-05 cm/s
C. 2.0e-04 cm/s
D. 5.0e-06 cm/s

Given equations / data

> $k_{v,eq} = \Sigma H_i / \Sigma(H_i/k_i)$; $k_{h,eq} = \Sigma(k_i H_i)/\Sigma H_i$

Worked solution

1. $\Sigma\ H_i/k_i = 4/2e\text{-}3 + 2/5e\text{-}6 + 6/8e\text{-}4 = 409{,}500$ ft·s/cm
2. $k_v = 12/409{,}500 = 2.93e\text{-}05$ cm/s
3. ($k_h = 1.07e\text{-}03$ cm/s for comparison; clay layer controls vertical flow)

Final answer

2.9e-05 cm/s

Why the other values are wrong

- **1.1e-03 cm/s** — used the horizontal (arithmetic) average
- **2.0e-04 cm/s** — used a geometric mean
- **5.0e-06 cm/s** — took the clay k alone

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.16.3 Flow Through Soil (layered/equivalent k), p. 188

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=RqaahrVtV4k (no creator values copied).

Personal use only — draft study material for Jayce Wright. Not for redistribution. Not reviewed by a licensed PE.